

- 12 (a)** (in a solid electrons in) neighbouring atoms are close together
(and influence/interact with each other) M1
- this changes their electron energy levels M1
- (many atoms in lattice) cause a spread of energy levels into a band A1 [3]

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- (b) photons of light give energy to electrons in valence band B1
- electrons move into the conduction band M1
- leaving holes in the valence band A1
- these electrons and holes are charge carriers B1
- increased number/increased current, hence reduced resistance B1 [5]